

Katlego Adult Learning Centre

From design to a working, tested network in Cisco Packet Tracer

Milestone 2: Client Implementation Review

Student	Konyana Lethabo
Student number	46967494
Project ID	CMPG325-2026-025
Client ID	CLI-025
Assigned feature	Port Security
Submission date	02 October 2026

Contents

1 The brief	2
1.1 What the client needs	2
1.2 What this milestone delivers	2
2 The design	2
2.1 The address plan	2
2.2 The equipment	3
2.3 Three decisions the build forced	3
3 Building the network	3
3.1 Layer 2: VLANs and trunks	4
3.2 Layer 3: routing between VLANs	5
3.3 Addresses: DHCP	6
3.4 Voice on the reception desk	6
4 Securing the network	6
4.1 Port Security (the assigned feature)	6
4.2 The server-room access list	7
5 Proving it works	8
5.1 Results at a glance	8
5.2 Reach: can devices talk to each other?	8
5.3 Restrict: who is kept out?	10
5.4 React: what happens when an unknown device appears?	12
5.4.1 Port Security across the whole network	13
5.5 Serve: do devices get what they need?	15
5.6 Configuration checks	16
6 What went wrong, and what changed	18
6.1 Where the build differs from the plan	18
7 The outcome	18

1 The brief

Katlego Adult Learning Centre needs a network that keeps its classrooms, computer lab, admin office and server room apart, and that keeps working without an IT department on site.

Katlego is a small institution with no dedicated IT staff on site. Milestone 1 turned its requirements into a design: five separate segments, one router, and a clear hierarchy of switches. Milestone 2 asks one question: **does that design work once it is built, and can we prove it?**

This report is the answer. It follows the order in which the network was built, and it ends with the evidence.

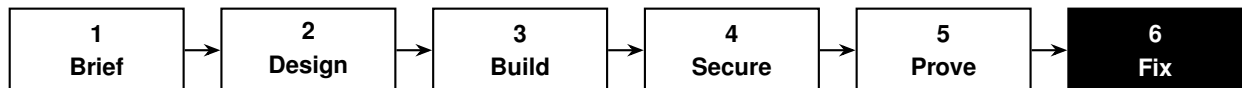


Figure 1. The storyline of this report: each section answers the question the previous one raised

1.1 What the client needs

Need	Meaning for the network	Met in
FR2	Separate VLANs for each part of the centre	Sections 3.1, 3.2
FR4	Port Security on every access port (the assigned technical challenge)	Sections 4.1, 5.4
FR5	Server room and infrastructure reachable by authorised staff only	Sections 4.2, 5.3
FR6 / CR11	VoIP handsets on their own segment	Sections 3.4, 5.5

1.2 What this milestone delivers

1. A working Packet Tracer file, 46967494_MILESTONE_2.pkt.
2. The assigned feature, Port Security, working on every access port.
3. Testing evidence: command output and connectivity tests.
4. An updated GitHub portfolio:
<https://github.com/konyanaethabo-work/katlego-network-project>

2 The design

Before any switch was touched, the Milestone 1 design was fixed: five segments, one router, one trunk.

Everything in this section was accepted in Milestone 1 and was not changed during the build.

2.1 The address plan

Each subnet is sized to the segment it serves. Management and the server room share VLAN 99.

VLAN	Segment	Network	Usable	Gateway	Addressing
10	Classrooms and Library	192.168.22.32/28	.33 – .46	192.168.22.33	DHCP
20	Computer Lab	192.168.22.0/27	.1 – .30	192.168.22.1	DHCP
30	Admin and Reception	192.168.22.48/28	.49 – .62	192.168.22.49	DHCP
40	Voice	192.168.22.64/29	.65 – .70	192.168.22.65	DHCP
99	Management, server room	192.168.22.72/29	.73 – .78	192.168.22.73	Static

2.2 The equipment

Role	Model	Why
Router	Cisco 1941	Sub-interfaces, ACLs and DHCP on one interface
Core, SW1, SW2	Cisco 2960-24TT	Layer 2 switching; no PoE needed
SW3 (Admin)	Catalyst 3560-24PS	PoE, so it can power the IP phone
IP phone	Cisco 7960	Standard Packet Tracer VoIP endpoint
End devices	19 PCs, 1 printer	Milestone 1 inventory

2.3 Three decisions the build forced

Milestone 1 fixed what the segments are. Building them forced three further decisions. Each was made once and applied everywhere.

Decision 1: how devices get their addresses

User VLANs use DHCP, because with no IT staff on site nobody should be typing addresses into twenty devices. The printer is the exception: it has a fixed address (192.168.22.50) because other devices print to it by address. VLAN 99 is fully static, so that management never depends on a DHCP service that is itself reached through the network it manages.

Decision 2: how Port Security reacts

Each access port allows only the devices it should carry: one learned (sticky) MAC address, or two on the phone port. A violation shuts the port down. With nobody watching the network, a port that quietly drops traffic goes unnoticed; a port that visibly goes down gets fixed. It also gives unambiguous evidence.

Decision 3: how the server room is protected

Milestone 1 gave the server room its own subnet. But with routing enabled, every VLAN could still reach it. An access list now blocks VLANs 10, 20 and 40 from 192.168.22.72/29. **Assumption:** the centre has no IT staff on site, so the Admin Office (VLAN 30) stands in for authorised staff and is deliberately left unrestricted. This extends Milestone 1; it did not state it.

3 Building the network

The network was built from the bottom up, because each layer depends on the one below.

First the switches learn the VLANs, then the router connects them, then devices receive addresses, then the phone is added. Every configuration below was typed into the device named above it.

3.1 Layer 2: VLANs and trunks

Each access switch knows only the VLANs it actually carries, plus VLAN 99 for management. Its uplink trunk is pruned to the same list, so broadcast traffic for a VLAN never reaches a switch with no devices in it. Only the router link carries all five.

KATLEGO-SW1-CLASSLIB

```
hostname KATLEGO-SW1-CLASSLIB
vlan 10
  name DATA_CLASSROOMS_LIBRARY
vlan 99
  name MANAGEMENT
interface vlan 99
  ip address 192.168.22.75 255.255.255.248
  no shutdown
ip default-gateway 192.168.22.73
interface range fastEthernet0/1 - 4
  switchport mode access
  switchport access vlan 10
interface fastEthernet0/24
  switchport mode trunk
  switchport trunk allowed vlan 10,99
```

KATLEGO-SW2-LAB

```
hostname KATLEGO-SW2-LAB
vlan 20
  name DATA_COMPUTER_LAB
vlan 99
  name MANAGEMENT
interface vlan 99
  ip address 192.168.22.76 255.255.255.248
  no shutdown
ip default-gateway 192.168.22.73
interface range fastEthernet0/1 - 11
  switchport mode access
  switchport access vlan 20
interface fastEthernet0/24
  switchport mode trunk
  switchport trunk allowed vlan 20,99
```

KATLEGO-SW3-ADMIN

```
hostname KATLEGO-SW3-ADMIN
vlan 30
  name DATA_ADMIN
vlan 40
  name VOICE
vlan 99
  name MANAGEMENT
interface vlan 99
  ip address 192.168.22.77 255.255.255.248
  no shutdown
ip default-gateway 192.168.22.73
interface range fastEthernet0/1 - 5
  switchport mode access
```

```
switchport access vlan 30
interface fastEthernet0/6
switchport mode access
switchport access vlan 30
switchport voice vlan 40
interface fastEthernet0/24
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 30,40,99
```

KATLEGO-CORE

```
hostname KATLEGO-CORE
vlan 10
name DATA_CLASSROOMS_LIBRARY
vlan 20
name DATA_COMPUTER_LAB
vlan 30
name DATA_ADMIN
vlan 40
name VOICE
vlan 99
name MANAGEMENT
interface vlan 99
ip address 192.168.22.74 255.255.255.248
no shutdown
ip default-gateway 192.168.22.73
interface fastEthernet0/1
switchport mode trunk
switchport trunk allowed vlan 10,99
interface fastEthernet0/2
switchport mode trunk
switchport trunk allowed vlan 20,99
interface fastEthernet0/4
switchport mode trunk
switchport trunk allowed vlan 30,40,99
interface gigabitEthernet0/1
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40,99
```

3.2 Layer 3: routing between VLANs

Switches keep VLANs apart; a router lets them talk. One physical link carries tagged traffic, and one sub-interface per VLAN acts as that VLAN's gateway.

```
hostname KATLEGO-RTR
interface gigabitEthernet0/0
no shutdown
interface gigabitEthernet0/0.10
encapsulation dot1Q 10
ip address 192.168.22.33 255.255.255.240
interface gigabitEthernet0/0.20
encapsulation dot1Q 20
ip address 192.168.22.1 255.255.255.224
interface gigabitEthernet0/0.30
encapsulation dot1Q 30
ip address 192.168.22.49 255.255.255.240
interface gigabitEthernet0/0.40
encapsulation dot1Q 40
ip address 192.168.22.65 255.255.255.248
```

```
interface gigabitEthernet0/0.99
  encapsulation dot1Q 99
  ip address 192.168.22.73 255.255.255.248
```

3.3 Addresses: DHCP

Gateways and the printer are excluded from the pools so they can never be handed out. There is no pool for VLAN 99, which is static by Decision 1.

```
ip dhcp excluded-address 192.168.22.1
ip dhcp excluded-address 192.168.22.33
ip dhcp excluded-address 192.168.22.49 192.168.22.50
ip dhcp excluded-address 192.168.22.65
ip dhcp pool LAB_VLAN20
  network 192.168.22.0 255.255.255.224
  default-router 192.168.22.1
  dns-server 8.8.8.8
ip dhcp pool CLASSLIB_VLAN10
  network 192.168.22.32 255.255.255.240
  default-router 192.168.22.33
  dns-server 8.8.8.8
ip dhcp pool ADMIN_VLAN30
  network 192.168.22.48 255.255.255.240
  default-router 192.168.22.49
  dns-server 8.8.8.8
ip dhcp pool VOICE_VLAN40
  network 192.168.22.64 255.255.255.248
  default-router 192.168.22.65
  option 150 ip 192.168.22.65
```

The printer is set by hand: 192.168.22.50, mask 255.255.255.240, gateway 192.168.22.49.

3.4 Voice on the reception desk

The phone sits on Fa0/6 of the PoE switch, which powers it. The port carries the data VLAN (30) and a voice VLAN (40), so the phone's traffic is tagged into the voice segment. Option 150 in the voice pool tells the phone where to find its call server. This satisfies FR6 and CR11.

In short. Switches know their VLANs, the router joins them, DHCP addresses the devices, and the phone has its own segment. The network works. It is not yet secure.

4 Securing the network

A working network is not a safe one. Two controls close the gaps: Port Security at the edge, and an access list at the server-room door.

4.1 Port Security (the assigned feature)

Port Security is applied to every access port, where a person can plug something in. Uplinks and core ports are excluded, because they connect switches, not people. Each port learns the first MAC address it sees and keeps it (sticky). If another address appears, the port shuts down.

Switch	Ports	Devices	Allowed MACs
SW1	Fa0/1 – Fa0/4	Classroom and library PCs	1 each
SW2	Fa0/1 – Fa0/11	Lab PCs	1 each
SW3	Fa0/1 – Fa0/3	Admin PCs	1 each
SW3	Fa0/4	Printer	1
SW3	Fa0/5	Reception PC	1
SW3	Fa0/6	IP phone	2

```

interface range fastEthernet0/1 - 5      ! single-device ports (SW3)
  switchport port-security
  switchport port-security maximum 1
  switchport port-security mac-address sticky
  switchport port-security violation shutdown
interface fastEthernet0/6                ! phone port (SW3)
  switchport port-security
  switchport port-security maximum 2
  switchport port-security mac-address sticky
  switchport port-security violation shutdown

```

SW1 and SW2 use the same four lines on their access ranges (Fa0/1 – 4 and Fa0/1 – 11), each with a maximum of 1.

4.2 The server-room access list

The list denies anything headed for 192.168.22.72/29 and permits everything else. It is applied inbound on the three VLANs it restricts.

```

access-list 100 remark Deny non-Admin VLANs from reaching the Server Room segment
access-list 100 deny ip any 192.168.22.72 0.0.0.7
access-list 100 permit ip any any
interface gigabitEthernet0/0.10
  ip access-group 100 in
interface gigabitEthernet0/0.20
  ip access-group 100 in
interface gigabitEthernet0/0.40
  ip access-group 100 in

```

The wildcard 0.0.0.7 covers exactly the VLAN 99 block, .72 to .79. The final permit is essential: an access list ends with an invisible deny, which would otherwise block all routing. It is not applied to VLAN 30, the deliberate exception, or to VLAN 99 itself.

In short. People can only plug in what the switch already knows, and only Admin can reach the server room. Now both claims have to be proved.

5 Proving it works

Every claim above is tested. The tests run in four steps: reach, restrict, react, serve.

5.1 Results at a glance

No.	Test	Observed	Result
1	Lab PC to Lab PC (same VLAN)	4 of 4 replies, TTL 128	Pass
2	Classroom PC to Lab PC (across VLANs)	4 of 4 replies, TTL 127 (one router hop)	Pass
3	Classroom and Lab PC to server room	Destination host unreachable; ACL counts the blocks	Pass
4	Admin PC to server room	4 of 4 replies, TTL 254	Pass
5	Unknown device on a secured port	Port err-disabled, violation counted, then recovered	Pass
6	DHCP leasing	20 leases, all in range, none excluded	Pass
7	Admin PC to IP phone	4 of 4 replies, TTL 254	Pass

5.2 Reach: can devices talk to each other?

Test 1: Connectivity within a VLAN	
Aim	Show that devices in the same VLAN communicate directly.
Method	PC4 (192.168.22.3) pinged another Lab PC, 192.168.22.2.
Expected	Replies with TTL 128: the packets never crossed the router.
Result	4 of 4 replies, 0% loss, TTL 128.

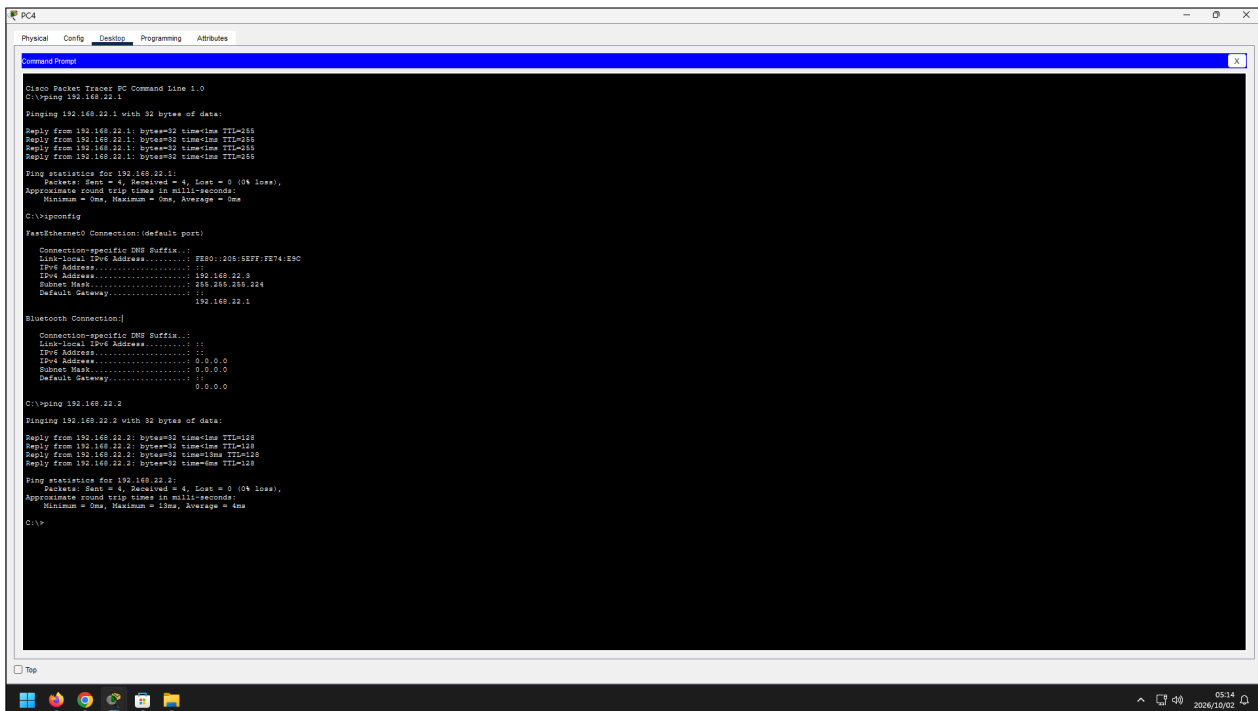


Figure 2. PC4 reaches its gateway and another Lab PC

Test 2: Connectivity between VLANs

Aim	Show that the router joins the VLANs.
Method	PC0 (VLAN 10) pinged PC4 (192.168.22.3, VLAN 20).
Expected	Replies with TTL 127: one router hop took 1 off 128.
Result	4 of 4 replies, 0% loss, TTL 127.

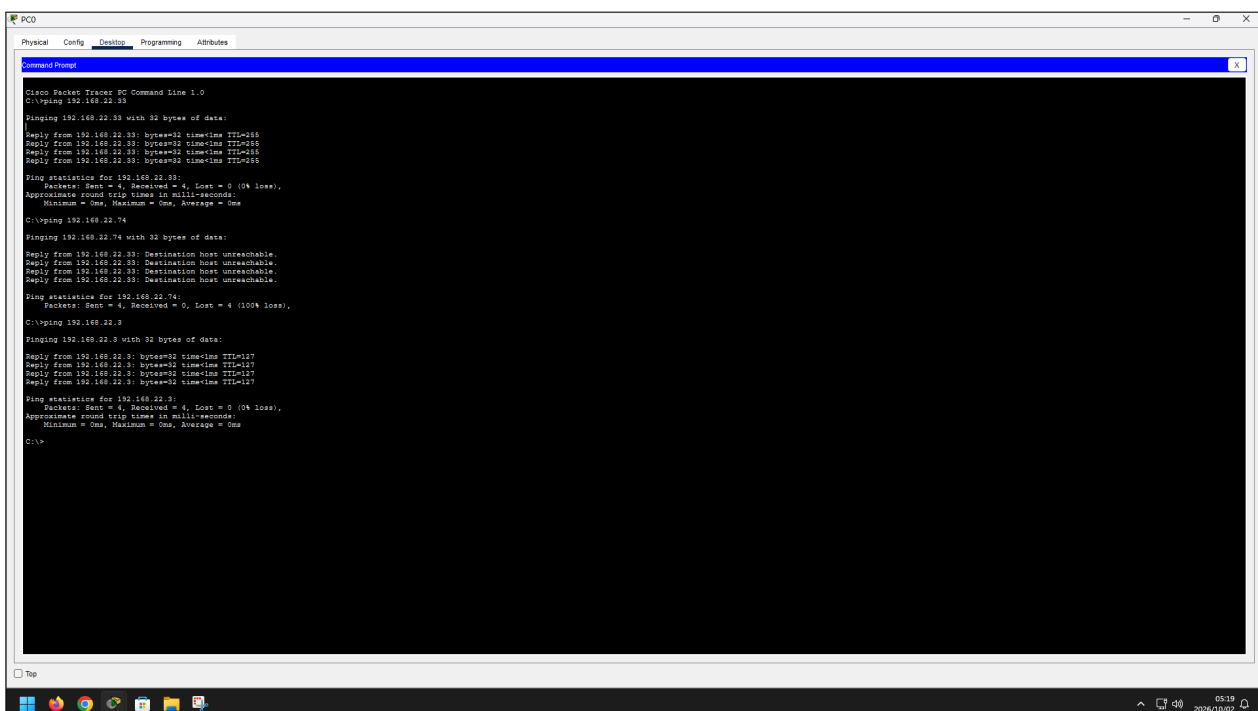


Figure 3. Classroom PC reaches a Lab PC through the router

5.3 Restrict: who is kept out?

Test 3: Restricted access to the server room (FR5)

Aim	Show that Classroom and Lab devices cannot reach the server-room segment.
Method	PC0 (VLAN 10) and PC4 (VLAN 20) each pinged the core switch at 192.168.22.74. Then show access-lists 100 was run on the router.
Expected	Both fail. The deny line counts the blocked packets.
Result	Destination host unreachable from the router for both; the deny entry shows 4 matches.

```

PC0
-----
Physical  Config  Desktop  Programming  Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.22.33

Pinging 192.168.22.33 with 32 bytes of data:

Reply from 192.168.22.33: bytes=32 time=1ms TTL=255
Reply from 192.168.22.33: bytes=32 time=1ms TTL=255
Reply from 192.168.22.33: bytes=32 time=1ms TTL=255
Reply from 192.168.22.33: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.22.33:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.22.74

Pinging 192.168.22.74 with 32 bytes of data:

Reply from 192.168.22.33: Destination host unreachable.
Reply from 192.168.22.33: Destination host unreachable.
Reply from 192.168.22.33: Destination host unreachable.
Reply from 192.168.22.33: Destination host unreachable.

Ping statistics for 192.168.22.74:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 192.168.22.3

Pinging 192.168.22.3 with 32 bytes of data:

Reply from 192.168.22.3: bytes=32 time=1ms TTL=127
Reply from 192.168.22.3: bytes=32 time=1ms TTL=127
Reply from 192.168.22.3: bytes=32 time=1ms TTL=127
Reply from 192.168.22.3: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.22.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
  
```

Figure 4. Classroom PC is blocked from the server-room segment

```

C:\>ping 192.168.22.74

Pinging 192.168.22.74 with 32 bytes of data:

Reply from 192.168.22.1: Destination host unreachable.
Reply from 192.168.22.1: Destination host unreachable.
Reply from 192.168.22.1: Destination host unreachable.
Reply from 192.168.22.1: Destination host unreachable.

Ping statistics for 192.168.22.74:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>
  
```

Figure 5. Lab PC is blocked from the server-room segment



Figure 6. The deny entry of ACL 100 counts the blocked packets

Test 4: The Admin exception

Aim	Show that Admin, and only Admin, can reach the server room.
Method	PC7 (VLAN 30) pinged 192.168.22.74.
Expected	Replies with TTL 254: the packets crossed the router and were allowed through.
Result	4 of 4 replies, 0% loss, TTL 254.

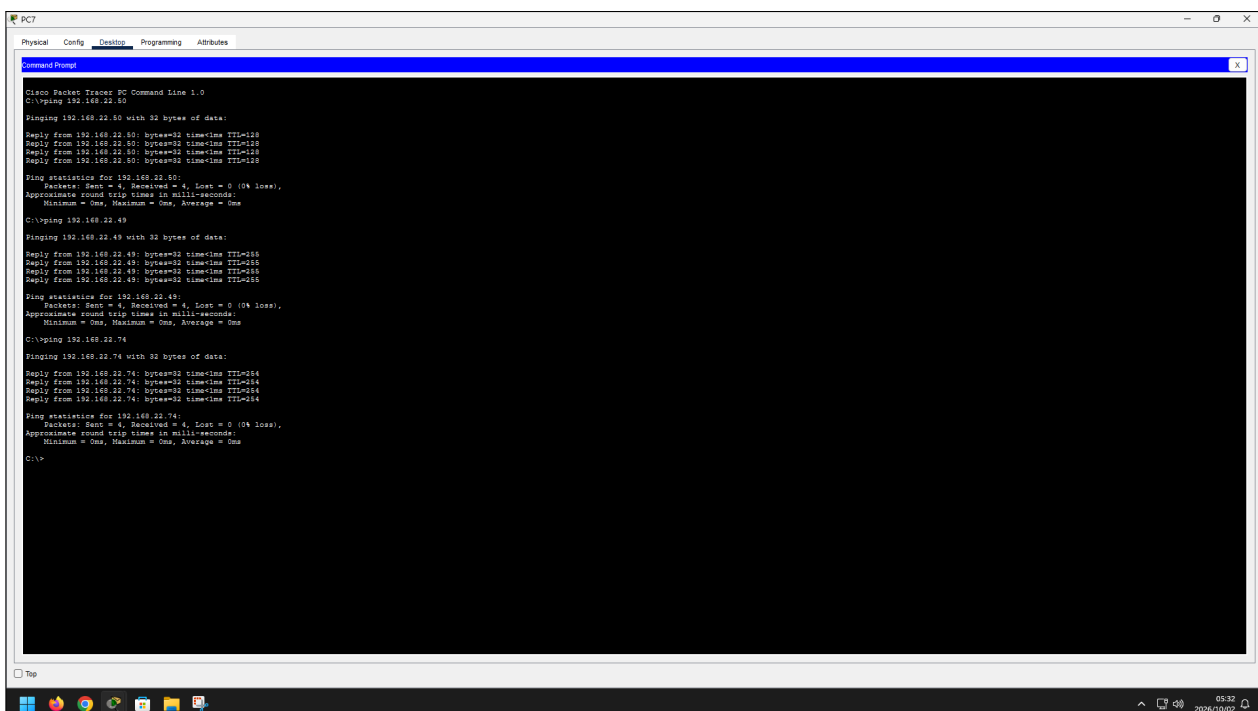


Figure 7. Admin PC reaches the server-room segment

5.4 React: what happens when an unknown device appears?

Test 5: Port Security violation (FR4)	
Aim	Show that a port shuts down when a device the switch has not learned is plugged in.
Method	On SW2 Fa0/4, PC4 was unplugged and a different PC connected and set to DHCP so that it sent traffic. Afterwards PC4 was reconnected and the port re-enabled with shutdown then no shutdown.
Expected	Violation logged, port err-disabled, other ports unaffected; after recovery PC4 is accepted again.
Result	Violation caused by MAC 0060.3E89.717E; Fa0/4 err-disabled, Secure-shutdown, violation count 1; recovered to Secure-up with PC4's MAC 0005.5E74.0E9C and count 0.

The four screenshots tell the story in order: before, the violation, the shutdown, the recovery. PC4's MAC was never re-learned, because the sticky entry survived the shutdown.

```

KATLEGO-SW2-LAB#show port-security interface fastEthernet0/4
Port Security          : Enabled
Port Status            : Secure-up
Violation Mode         : Shutdown
Aging Time             : 0 mins
Aging Type             : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses  : 1
Total MAC Addresses    : 1
Configured MAC Addresses : 0
Sticky MAC Addresses   : 1
Last Source Address:Vlan : 0005.5E74.0E9C:20
Security Violation Count : 0

KATLEGO-SW2-LAB#

```

Figure 8. Before: Fa0/4 is Secure-up with 0 violations

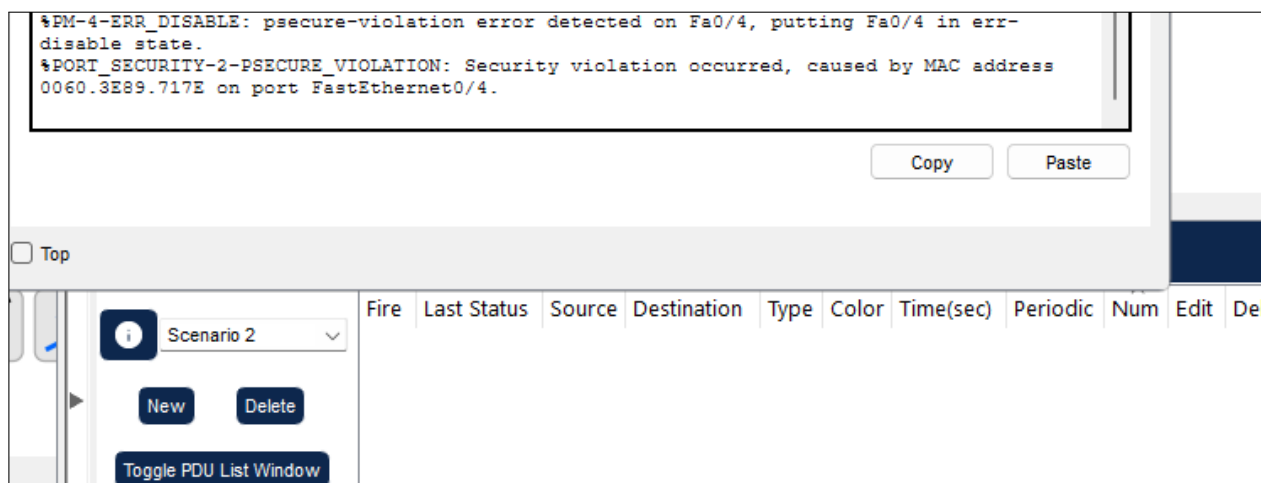


Figure 9. The violation: the switch logs the unknown MAC and err-disables the port

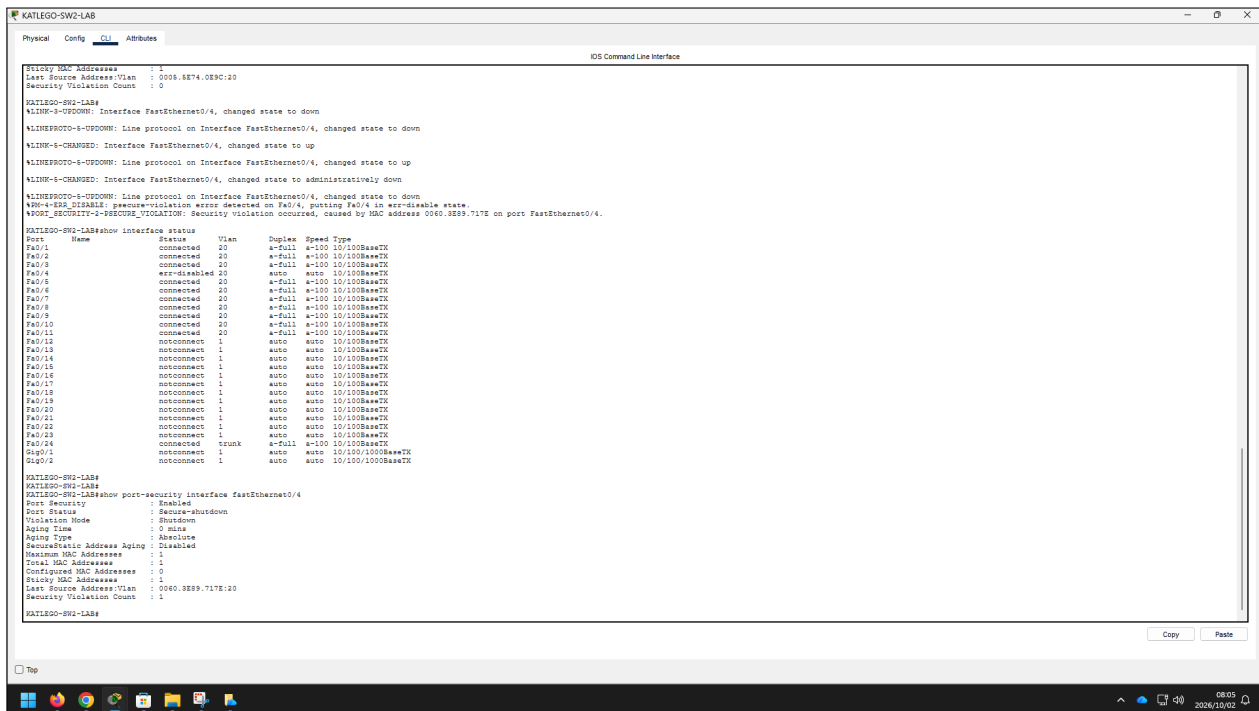


Figure 10. The shutdown: Fa0/4 is err-disabled, violation count 1, other Lab ports still connected

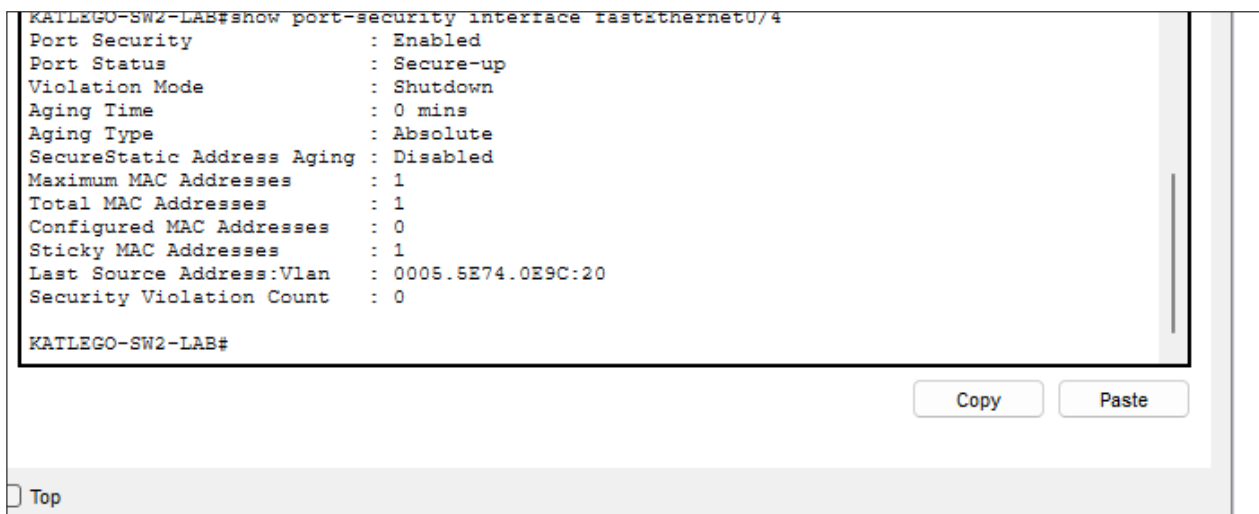


Figure 11. The recovery: Fa0/4 is Secure-up again with PC4's MAC retained

Port Security across the whole network

After the test, each switch confirms its limits, its learned addresses and its action. Every access port shows no violations.

```
KATLEGO-SW1-CLASSLIB>enable
KATLEGO-SW1-CLASSLIB#show port-security
```

Secure	Port	MaxSecureAddr (Count)	CurrentAddr (Count)	SecurityViolation (Count)	Security Action
	Fa0/1	1	1	0	Shutdown
	Fa0/2	1	1	0	Shutdown
	Fa0/3	1	1	0	Shutdown
	Fa0/4	1	1	0	Shutdown

Figure 12. SW1: Fa0/1 to Fa0/4, maximum 1, Shutdown

```
KATLEGO-SW2-LAB#show port-security
```

Secure	Port	MaxSecureAddr (Count)	CurrentAddr (Count)	SecurityViolation (Count)	Security Action
	Fa0/1	1	1	0	Shutdown
	Fa0/2	1	1	0	Shutdown
	Fa0/3	1	1	0	Shutdown
	Fa0/4	1	1	0	Shutdown
	Fa0/5	1	1	0	Shutdown
	Fa0/6	1	1	0	Shutdown
	Fa0/7	1	1	0	Shutdown
	Fa0/8	1	1	0	Shutdown
	Fa0/9	1	1	0	Shutdown
	Fa0/10	1	1	0	Shutdown
	Fa0/11	1	1	0	Shutdown

KATLEGO-SW2-LAB#

Copy Paste

Figure 13. SW2: Fa0/1 to Fa0/11, maximum 1, Shutdown

```
KATLEGO-SW3-ADMIN>enable
KATLEGO-SW3-ADMIN#show port-security
```

Secure	Port	MaxSecureAddr (Count)	CurrentAddr (Count)	SecurityViolation (Count)	Security Action
	Fa0/1	1	1	0	Shutdown
	Fa0/2	1	1	0	Shutdown
	Fa0/3	1	1	0	Shutdown
	Fa0/4	1	1	0	Shutdown
	Fa0/5	1	1	0	Shutdown
	Fa0/6	2	1	0	Shutdown

KATLEGO-SW3-ADMIN#

Copy Paste

Figure 14. SW3: Fa0/1 to Fa0/5 maximum 1, phone port Fa0/6 maximum 2

5.5 Serve: do devices get what they need?

Test 6: DHCP leasing

Aim	Show that every DHCP device receives a correct address and no reserved address is leased.
Method	show ip dhcp binding on the router; ipconfig on PC0.
Expected	Addresses inside each VLAN's range; none of .1, .33, .49, .50, .65.
Result	20 leases: 11 in the Lab, 4 in Classrooms and Library, 4 in Admin, 1 voice. PC0 received 192.168.22.37, mask 255.255.255.240, gateway 192.168.22.33.

KATLEGO-RTR#show ip dhcp binding			
IP address	Client-ID/ Hardware address	Lease expiration	Type
192.168.22.4	000C.CF64.E508	--	Automatic
192.168.22.2	00D0.D380.1A11	--	Automatic
192.168.22.5	0001.42C2.B6C9	--	Automatic
192.168.22.6	00E0.B04C.0CB7	--	Automatic
192.168.22.10	00D0.FF2A.7214	--	Automatic
192.168.22.11	0001.967C.7EBE	--	Automatic
192.168.22.9	0006.2A0A.8ADD	--	Automatic
192.168.22.8	0060.2F39.9357	--	Automatic
192.168.22.7	00D0.FFC4.66C6	--	Automatic
192.168.22.12	0000.0C10.16C2	--	Automatic
192.168.22.3	0005.5E74.0E9C	--	Automatic
192.168.22.34	0010.11C8.710C	--	Automatic
192.168.22.35	0060.7048.4BC2	--	Automatic
192.168.22.36	0002.16D4.4633	--	Automatic
192.168.22.37	0060.2FCA.9E9A	--	Automatic
192.168.22.51	0060.3EE6.7B5B	--	Automatic
192.168.22.52	0001.C7D1.36DA	--	Automatic
192.168.22.53	00E0.A370.638D	--	Automatic
192.168.22.54	000D.BDC6.7BA8	--	Automatic
192.168.22.66	0002.17CC.2640	--	Automatic
KATLEGO-RTR#			

Figure 15. DHCP bindings on the router

```

C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::260:2FFF:FECA:9E9A
    IPv6 Address . . . . .: ::
    IPv4 Address. . . . .: 192.168.22.37
    Subnet Mask . . . . .: 255.255.255.240
    Default Gateway . . . . .: ::
                                192.168.22.33

```

Figure 16. PC0 holds an address from the VLAN 10 pool

Test 7: VoIP phone (FR6 / CR11)

Aim	Show that the phone is powered, addressed in the voice VLAN, and reachable.
Method	The phone leased 192.168.22.66. PC7 in VLAN 30 pinged it.
Expected	Replies with TTL 254: the voice VLAN is reachable through the router.
Result	4 of 4 replies, 0% loss, TTL 254.

```

C:\>ping 192.168.22.66

Pinging 192.168.22.66 with 32 bytes of data:

Reply from 192.168.22.66: bytes=32 time<1ms TTL=254
Reply from 192.168.22.66: bytes=32 time<1ms TTL=254
Reply from 192.168.22.66: bytes=32 time<1ms TTL=254
Reply from 192.168.22.66: bytes=32 time<1ms TTL=254

Ping statistics for 192.168.22.66:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

```

☐ Top

Figure 17. Admin PC reaches the IP phone in the voice VLAN

5.6 Configuration checks

Three last outputs confirm the configuration itself, not just its behaviour.

```

KATLEGO-SW1-CLASSLIB>enable
KATLEGO-SW1-CLASSLIB#show vlan brief

VLAN Name                Status    Ports
----
1    default                active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                           Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                           Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                           Fa0/21, Fa0/22, Fa0/23, Gig0/1
                                           Gig0/2
10   DATA_CLASSROOMS_LIBRARY active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
99   MANAGEMENT              active

```

Figure 18. VLANs on SW1: VLAN 10 holds Fa0/1 to Fa0/4, VLAN 99 exists

```

KATLEGO-CORE>enable
KATLEGO-CORE#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     on        802.1q         trunking    1
Fa0/2     on        802.1q         trunking    1
Fa0/4     on        802.1q         trunking    1
Gig0/1    on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/1     10,99
Fa0/2     20,99
Fa0/4     30,40,99
Gig0/1    10,20,30,40,99

Port      Vlans allowed and active in management domain
Fa0/1     10,99
Fa0/2     20,99
Fa0/4     30,40,99
Gig0/1    10,20,30,40,99

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     10,99
Fa0/2     20,99
Fa0/4     30,40,99
Gig0/1    10,20,30,40,99

KATLEGO-CORE#
KATLEGO-CORE#
KATLEGO-CORE##

```

Copy Paste

Figure 19. Core trunks: each carries only its pruned VLAN list

```

KATLEGO-RTR#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0    unassigned      YES NVRAM   up          up
GigabitEthernet0/0.10 192.168.22.33   YES manual up          up
GigabitEthernet0/0.20 192.168.22.1    YES manual up          up
GigabitEthernet0/0.30 192.168.22.49   YES manual up          up
GigabitEthernet0/0.40 192.168.22.65   YES manual up          up
GigabitEthernet0/0.99 192.168.22.73   YES manual up          up

```

Figure 20. Router: Gi0/0 and five sub-interfaces up and addressed

In short. All seven tests passed. The network reaches where it should, refuses where it should, shuts down a rogue device, and serves its users. But it did not pass first time.

6 What went wrong, and what changed

The first build did not pass every test. Six problems turned up. Each was found by testing and fixed at its cause.

No.	What we saw	Cause	Fix
1	A DHCP address conflict logged for 192.168.22.33	Exclusion range typed with its end below its start, leaving the VLAN 10 gateway unprotected	Removed it; added a single <code>excluded-address 192.168.22.33</code>
2	Voice pool incomplete	<code>option 150</code> missing	Added <code>option 150 ip 192.168.22.65</code>
3	SW3 trunk on the core carrying every VLAN	Core port left to negotiate, so it was unpruned	Set it as a fixed trunk allowing only VLANs 30, 40, 99
4	Phone port err-disabled at start-up	Two different MAC addresses seen on the port, with a maximum of 1	Maximum set to 2 on that port
5	Phone port blocked by spanning tree; phone did not answer pings	Port was a plain VLAN 40 access port with no voice VLAN	Access VLAN 30 with <code>switchport voice vlan 40</code> ; phone then answered
6	ACL missing from the voice VLAN	Applied only to VLANs 10 and 20	Applied to <code>Gi0/0.40</code>

6.1 Where the build differs from the plan

- **SW3 model:** a Catalyst 3560-24PS replaced the planned 2960-24PC. It is PoE-capable and powers the phone, so the requirement is met.
- **Phone and reception PC:** they sit on separate ports, Fa0/6 (maximum 2 MACs) and Fa0/5 (maximum 1), rather than sharing one.
- **SW3 uplink:** it connects to core Fa0/4; core Fa0/3 is spare.
- **Device count:** 19 PCs, the printer and the phone make 21 devices.

7 The outcome

The Katlego network works as designed, keeps its segments apart, protects its edge, and can prove it.

Five VLANs, router-on-a-stick routing, DHCP, a voice segment, Port Security on every access port and a server-room access list were configured and tested. All seven tests pass, and the one assigned feature, Port Security, was demonstrated by causing a real violation and recovering from it. The problems found on the way are recorded above as part of that evidence.

Milestone 2 deliverables

1. Packet Tracer file: 46967494_MILESTONE_2.pkt
2. This report, with the testing evidence
3. GitHub portfolio:
<https://github.com/konyanaethabo-work/katlego-network-project>